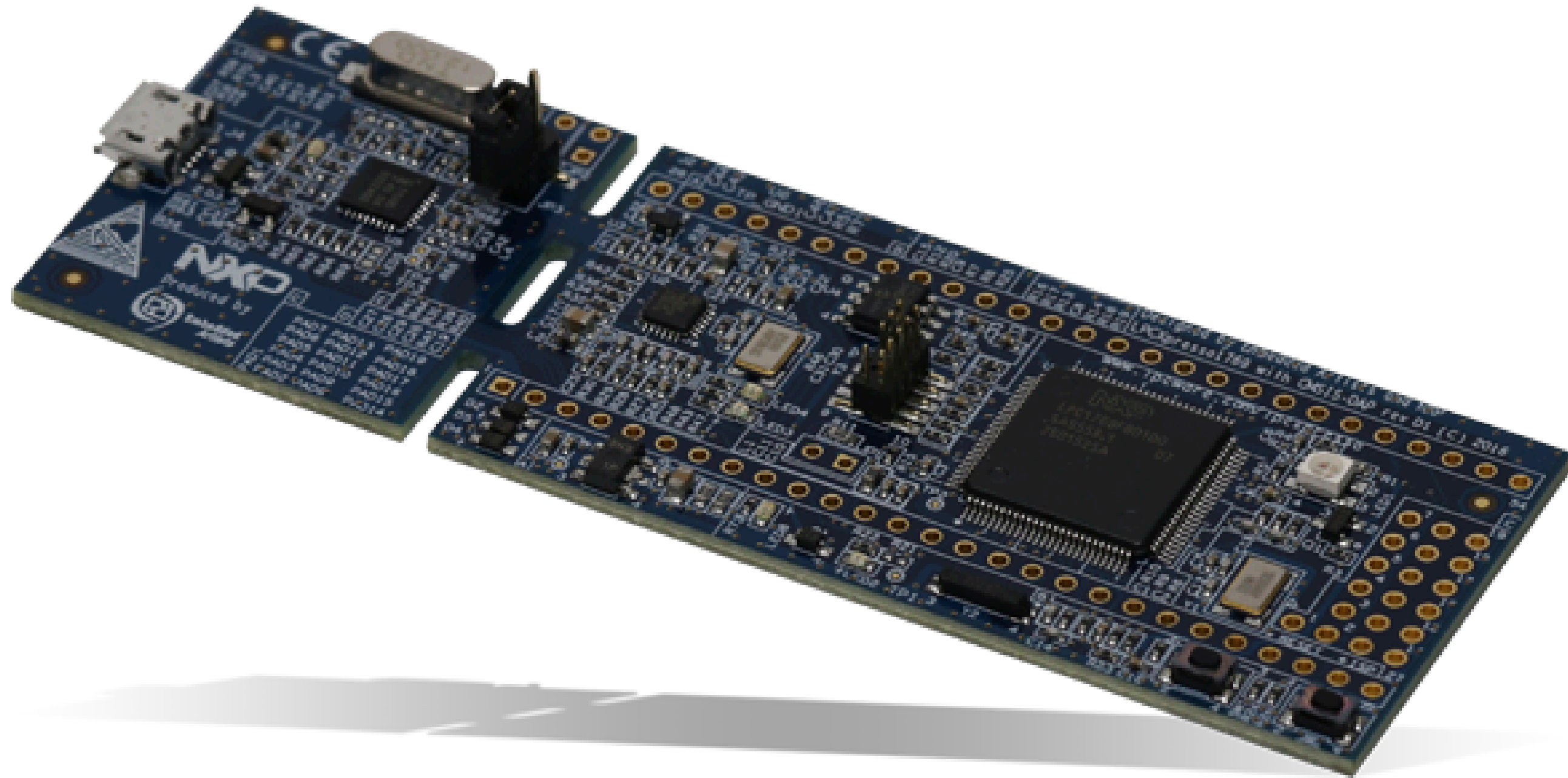




UART

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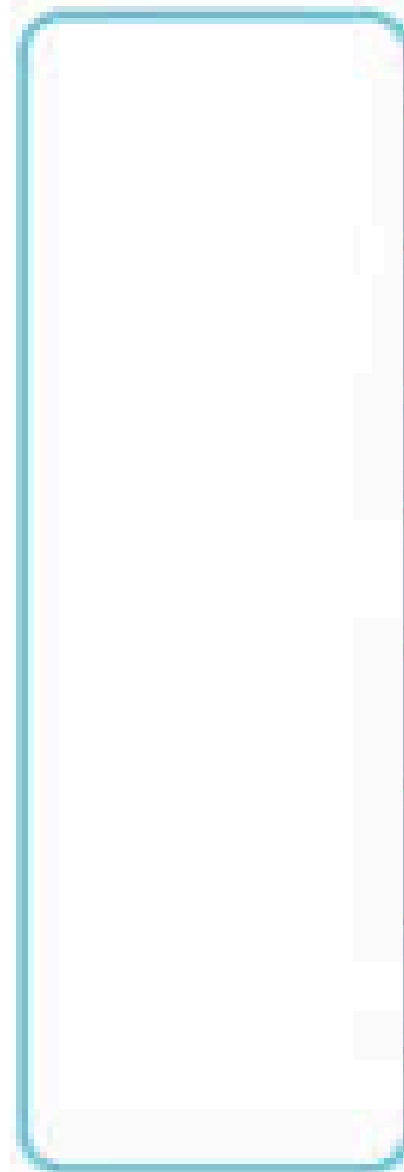


COMMUNICATION MODES

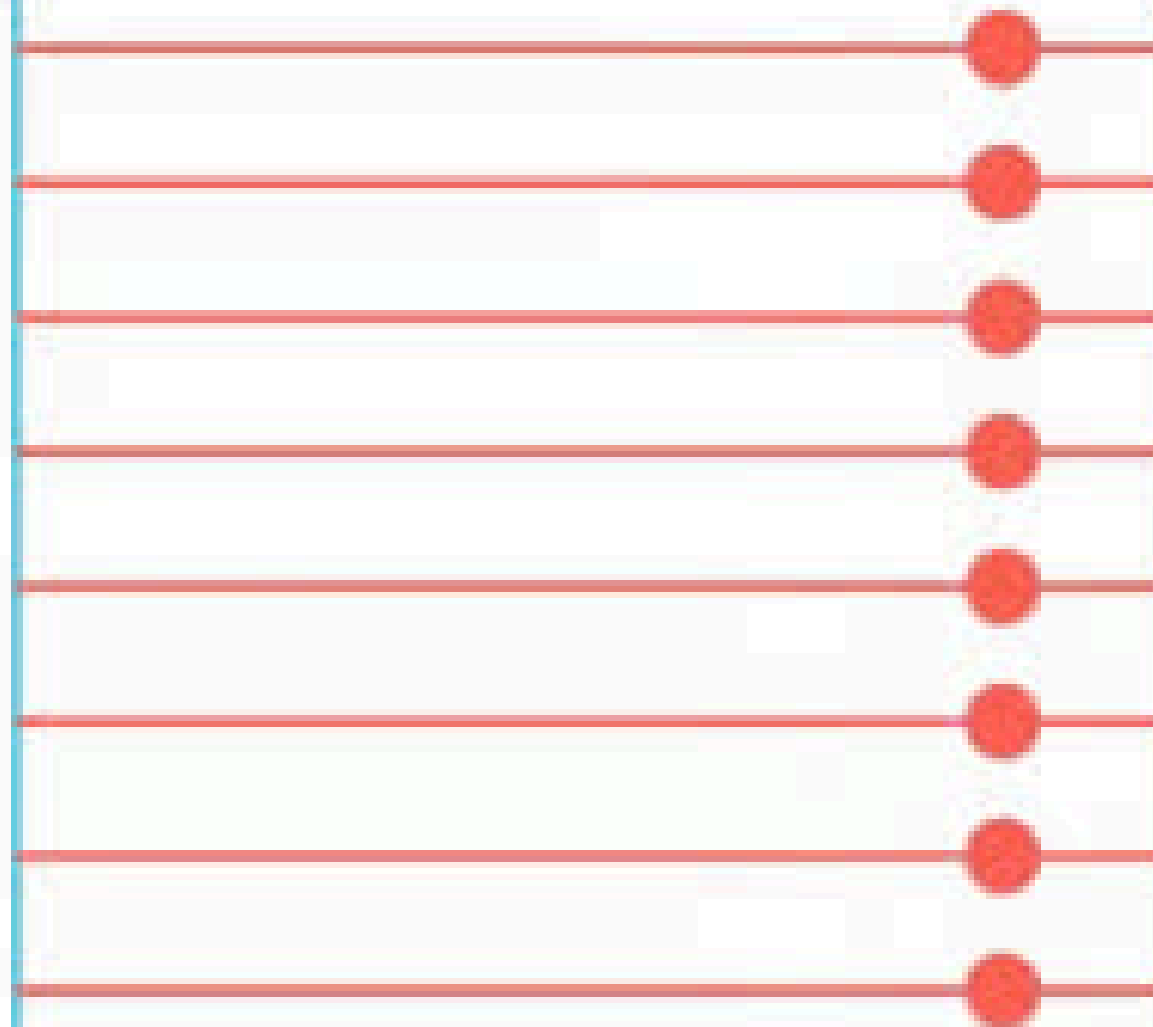
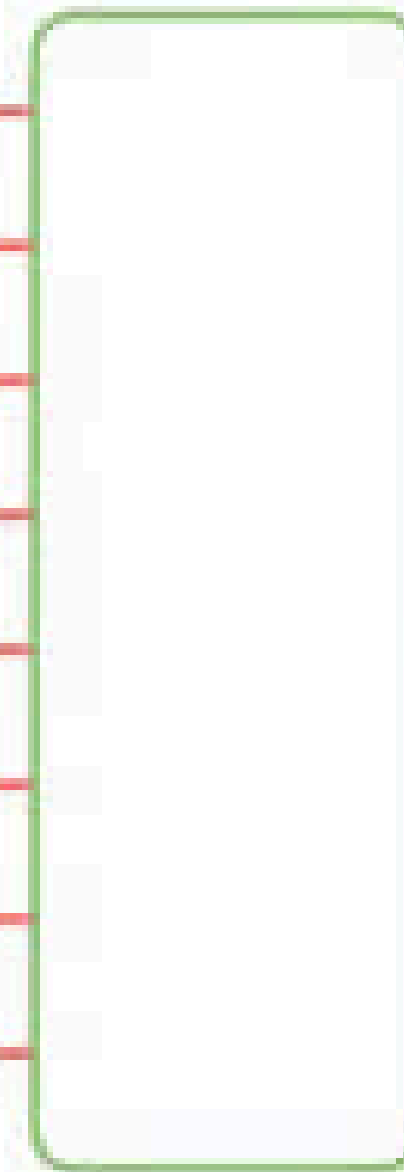
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Parallel Communication

Device A



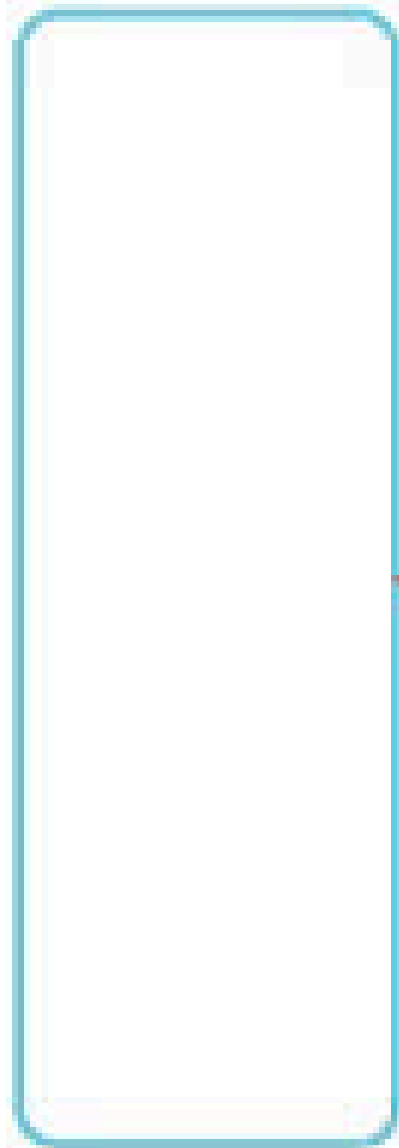
Device A



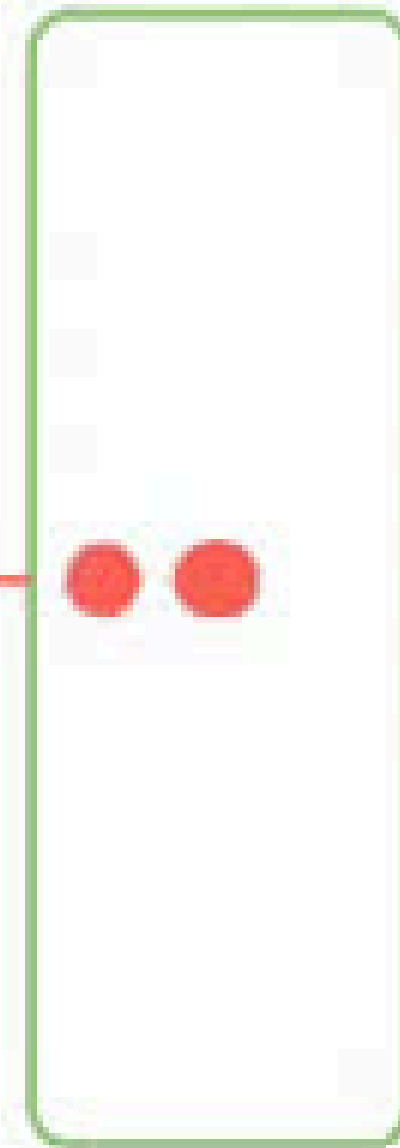
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Serial Communication

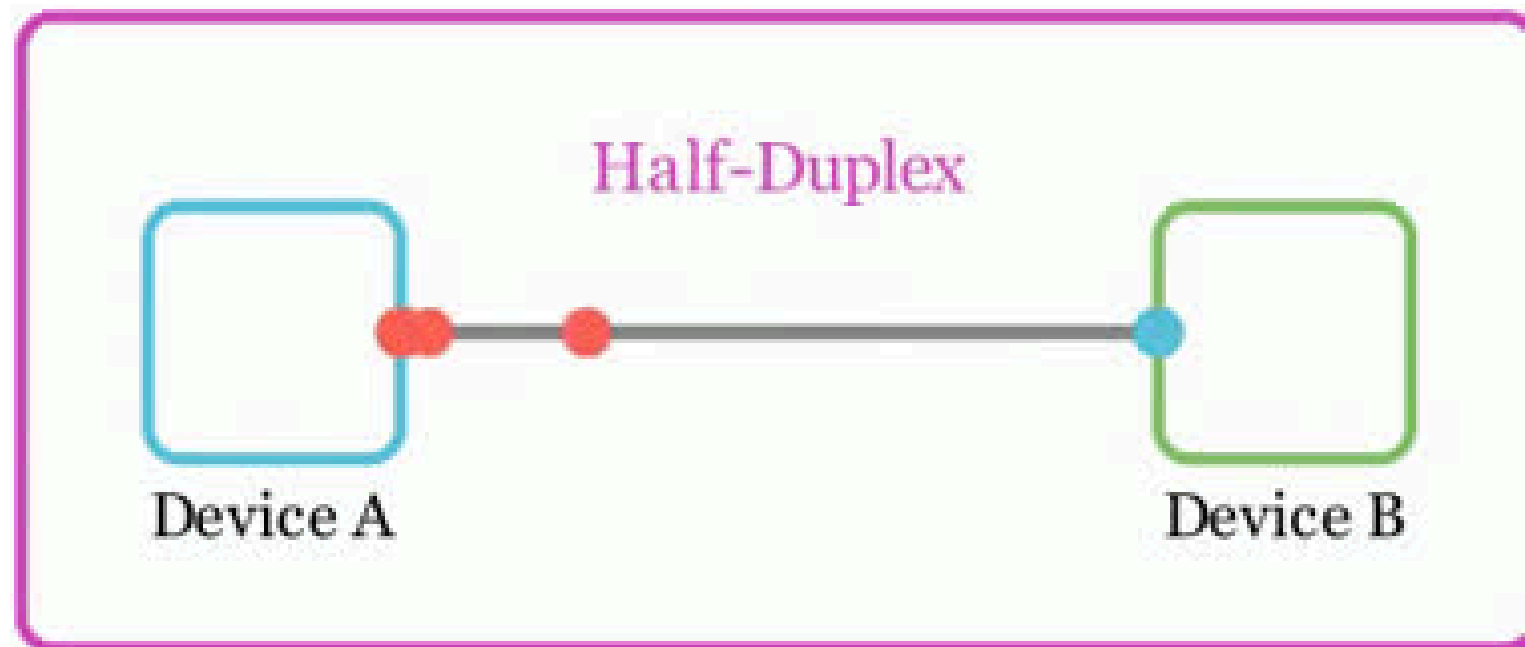
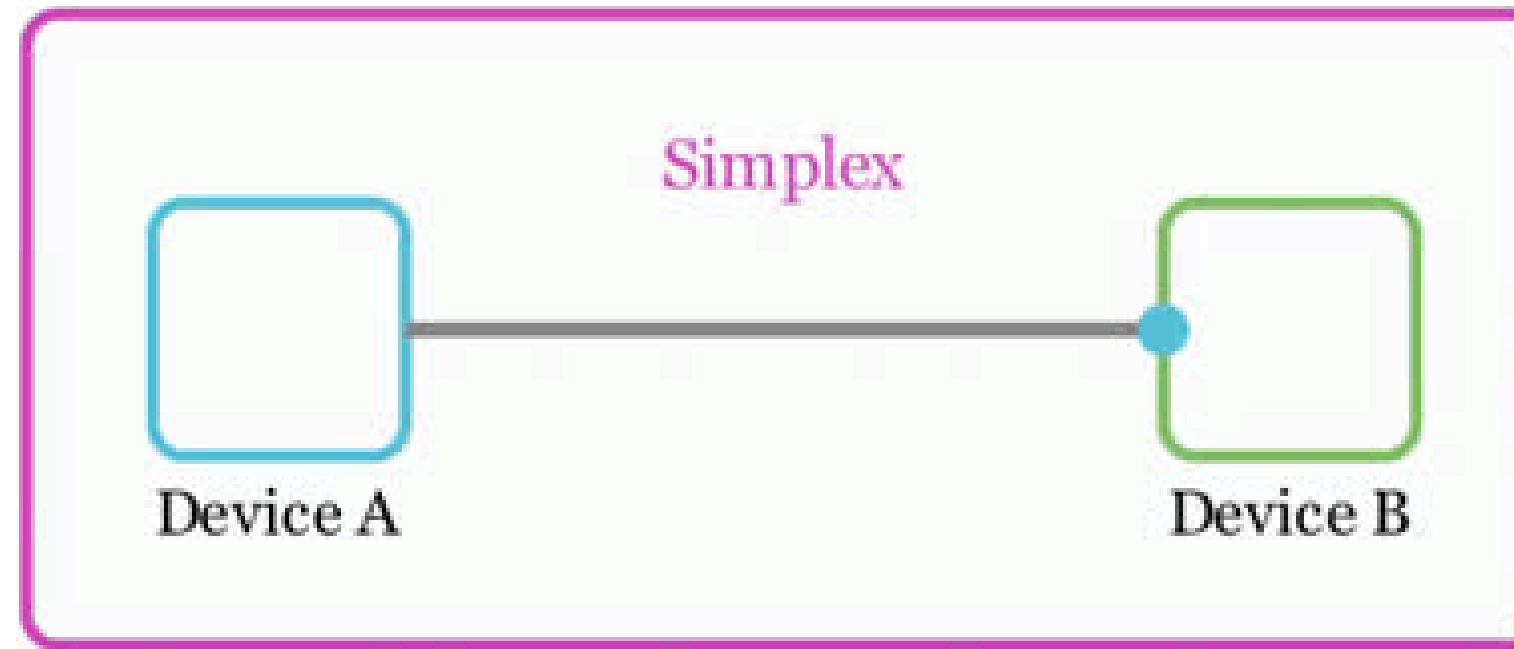
Device A



Device A



David Trujillo



David Trujillo

14.2 Features

- Data sizes of 5, 6, 7, and 8 bits.
- Parity generation and checking: odd, even mark, space or none.
- One or two stop bits.
- 16 byte Receive and Transmit FIFOs.
- Built-in baud rate generator, including a fractional rate divider for great versatility.
- Supports DMA for both transmit and receive.
- Auto-baud capability
- Break generation and detection.
- Multiprocessor addressing mode.
- IrDA mode to support infrared communication.
- Support for software flow control.

AVAILABLE PINS

TXD0	P0.2	01		
RXD0	P0.3	01		
TXD1	P0.15	01	P2.0	10
RXD1	P0.16	01	P2.1	10
TXD2	P0.10	01	P2.8	10
RXD2	P0.11	01	P2.9	10
TXD3	P0.0	10	P4.28	11
RXD3	P0.1	10	P4.29	11

**CLOCK
CONTROL**

Table 46. Power Control for Peripherals register (PCONP - address 0x400F C0C4) bit description

Bit	Symbol	Description	Reset value
3	PCUART0	UART0 power/clock control bit.	1
4	PCUART1	UART1 power/clock control bit.	1
● ● ●			
24	PCUART2	UART 2 power/clock control bit.	0
25	PCUART3	UART 3 power/clock control bit.	0

Table 40. Peripheral Clock Selection register 0 (PCLKSEL0 - address 0x400F C1A8) bit description

Bit	Symbol	Description	Reset value
7:6	PCLK_UART0	Peripheral clock selection for UART0.	00
9:8	PCLK_UART1	Peripheral clock selection for UART1.	00

Table 41. Peripheral Clock Selection register 1 (PCLKSEL1 - address 0x400F C1AC) bit description

Bit	Symbol	Description	Reset value
17:16	PCLK_UART2	Peripheral clock selection for UART2.	00
19:18	PCLK_UART3	Peripheral clock selection for UART3.	00

Table 42. Peripheral Clock Selection register bit values

PCLKSEL0 and PCLKSEL1 individual peripheral's clock select options	Function	Reset value
00	PCLK_peripheral = CCLK/4	00
01	PCLK_peripheral = CCLK	
10	PCLK_peripheral = CCLK/2	
11	PCLK_peripheral = CCLK/8, except for CAN1, CAN2, and CAN filtering when "11" selects = CCLK/6.	

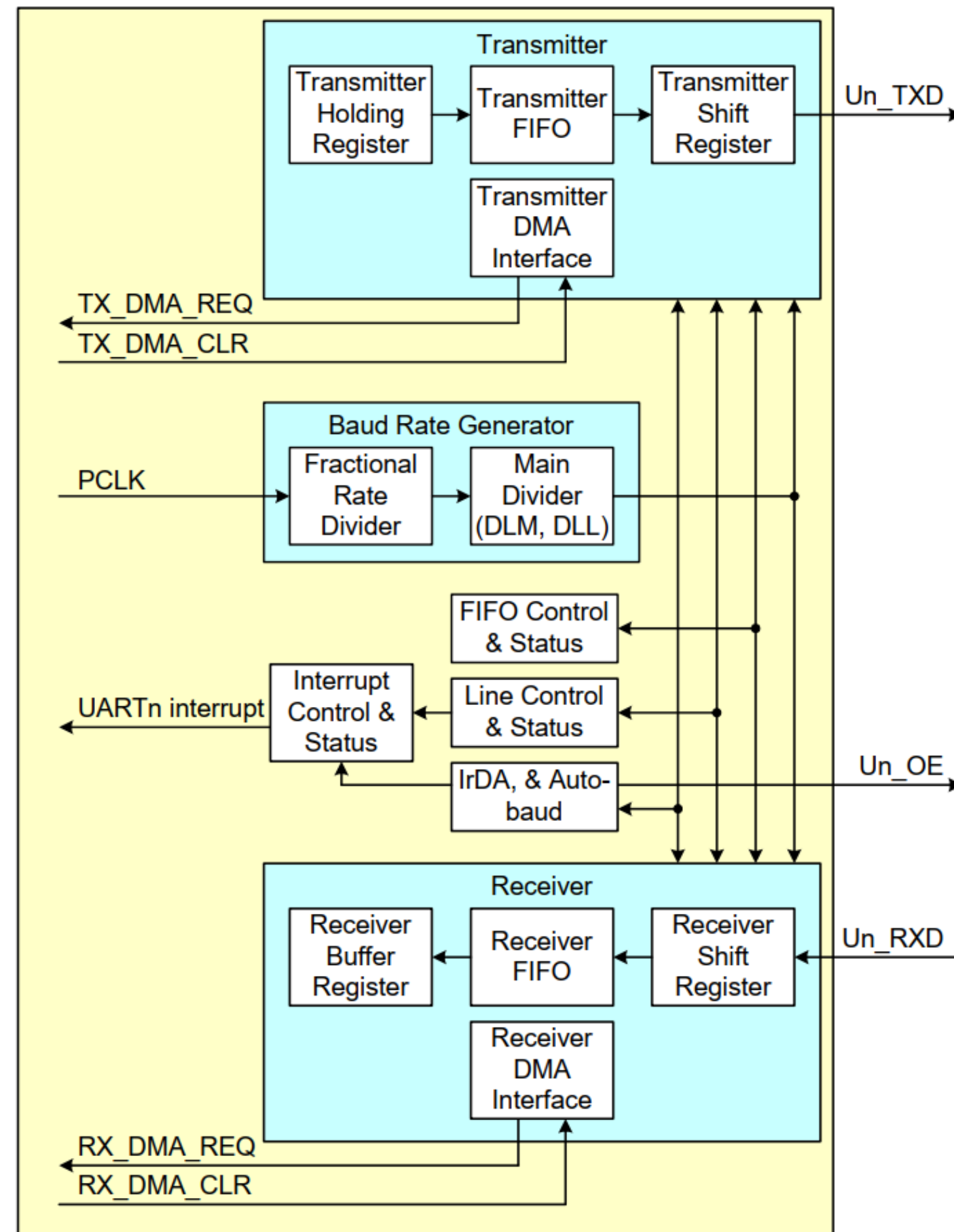


Fig 48. UART0, 2 and 3 block diagram

Table 271. UART0/2/3 Register Map

Generic Name	Description	Access	Reset value ^[1]	UARTn Register Name & Address
RBR (DLAB =0)	Receiver Buffer Register. Contains the next received character to be read.	RO	NA	U0RBR - 0x4000 C000 U2RBR - 0x4009 8000 U3RBR - 0x4009 C000
THR (DLAB =0)	Transmit Holding Register. The next character to be transmitted is written here.	WO	NA	U0THR - 0x4000 C000 U2THR - 0x4009 8000 U3THR - 0x4009 C000
DLL (DLAB =1)	Divisor Latch LSB. Least significant byte of the baud rate divisor value. The full divisor is used to generate a baud rate from the fractional rate divider.	R/W	0x01	U0DLL - 0x4000 C000 U2DLL - 0x4009 8000 U3DLL - 0x4009 C000
DLM (DLAB =1)	Divisor Latch MSB. Most significant byte of the baud rate divisor value. The full divisor is used to generate a baud rate from the fractional rate divider.	R/W	0x00	U0DLM - 0x4000 C004 U2DLM - 0x4009 8004 U3DLM - 0x4009 C004
IER (DLAB =0)	Interrupt Enable Register. Contains individual interrupt enable bits for the 7 potential UART interrupts.	R/W	0x00	U0IER - 0x4000 C004 U2IER - 0x4009 8004 U3IER - 0x4009 C004
IIR	Interrupt ID Register. Identifies which interrupt(s) are pending.	RO	0x01	U0IIR - 0x4000 C008 U2IIR - 0x4009 8008 U3IIR - 0x4009 C008
FCR	FIFO Control Register. Controls UART FIFO usage and modes.	WO	0x00	U0FCR - 0x4000 C008 U2FCR - 0x4009 8008 U3FCR - 0x4009 C008

Table 271. UART0/2/3 Register Map

Generic Name	Description	Access	Reset value ^[1]	UARTn Register Name & Address
LCR	Line Control Register. Contains controls for frame formatting and break generation.	R/W	0x00	U0LCR - 0x4000 C00C U2LCR - 0x4009 800C U3LCR - 0x4009 C00C
LSR	Line Status Register. Contains flags for transmit and receive status, including line errors.	RO	0x60	U0LSR - 0x4000 C014 U2LSR - 0x4009 8014 U3LSR - 0x4009 C014
SCR	Scratch Pad Register. 8-bit temporary storage for software.	R/W	0x00	U0SCR - 0x4000 C01C U2SCR - 0x4009 801C U3SCR - 0x4009 C01C
ACR	Auto-baud Control Register. Contains controls for the auto-baud feature.	R/W	0x00	U0ACR - 0x4000 C020 U2ACR - 0x4009 8020 U3ACR - 0x4009 C020
ICR	IrDA Control Register. Enables and configures the IrDA mode.	R/W	0x00	U0ICR - 0x4000 C024 U2ICR - 0x4009 8024 U3ICR - 0x4009 C024
FDR	Fractional Divider Register. Generates a clock input for the baud rate divider.	R/W	0x10	U0FDR - 0x4000 C028 U2FDR - 0x4009 8028 U3FDR - 0x4009 C028
TER	Transmit Enable Register. Turns off UART transmitter for use with software flow control.	R/W	0x80	U0TER - 0x4000 C030 U2TER - 0x4009 8030 U3TER - 0x4009 C030

Table 280: UARTn Line Control Register (U0LCR - address 0x4000 C00C, U2LCR - 0x4009 800C, U3LCR - 0x4009 C00C) bit description

Bit	Symbol	Value	Description	Reset Value
1:0	Word Length Select	00	5-bit character length	0
		01	6-bit character length	
		10	7-bit character length	
		11	8-bit character length	
2	Stop Bit Select	0	1 stop bit.	0
		1	2 stop bits (1.5 if UnLCR[1:0]=00).	
3	Parity Enable	0	Disable parity generation and checking.	0
		1	Enable parity generation and checking.	
5:4	Parity Select	00	Odd parity. Number of 1s in the transmitted character and the attached parity bit will be odd.	0
		01	Even Parity. Number of 1s in the transmitted character and the attached parity bit will be even.	
		10	Forced "1" stick parity.	
		11	Forced "0" stick parity.	
6	Break Control	0	Disable break transmission.	0
		1	Enable break transmission. Output pin UARTn TXD is forced to logic 0 when UnLCR[6] is active high.	
7	Divisor Latch Access Bit (DLAB)	0	Disable access to Divisor Latches.	0
		1	Enable access to Divisor Latches.	
31:8	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

Table 274: UARTn Divisor Latch LSB register (U0DLL - address 0x4000 C000, U2DLL - 0x4009 8000, U3DLL - 0x4009 C000 when DLAB = 1) bit description

Bit	Symbol	Description	Reset Value
7:0	DLLSB	The UARTn Divisor Latch LSB Register, along with the UnDLM register, determines the baud rate of the UARTn.	0x01
31:8	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

Table 275: UARTn Divisor Latch MSB register (U0DLM - address 0x4000 C004, U2DLM - 0x4009 8004, U3DLM - 0x4009 C004 when DLAB = 1) bit description

Bit	Symbol	Description	Reset Value
7:0	DLMSB	The UARTn Divisor Latch MSB Register, along with the U0DLL register, determines the baud rate of the UARTn.	0x00
31:8	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

Table 286: UARTn Fractional Divider Register (U0FDR - address 0x4000 C028, U2FDR - 0x4009 8028, U3FDR - 0x4009 C028) bit description

Bit	Function	Value	Description	Reset value
3:0	DIVADDVAL	0	Baud-rate generation pre-scaler divisor value. If this field is 0, fractional baud-rate generator will not impact the UARTn baudrate.	0
7:4	MULVAL	1	Baud-rate pre-scaler multiplier value. This field must be greater or equal 1 for UARTn to operate properly, regardless of whether the fractional baud-rate generator is used or not.	1
31:8	-	NA	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	0

$$UARTn_{baudrate} = \frac{PCLK}{16 \times (256 \times UnDLM + UnDLL) \times \left(1 + \frac{DivAddVal}{MulVal}\right)}$$

The value of MULVAL and DIVADDVAL should comply to the following conditions:

1. $1 \leq MULVAL \leq 15$
2. $0 \leq DIVADDVAL \leq 14$
3. $DIVADDVAL < MULVAL$

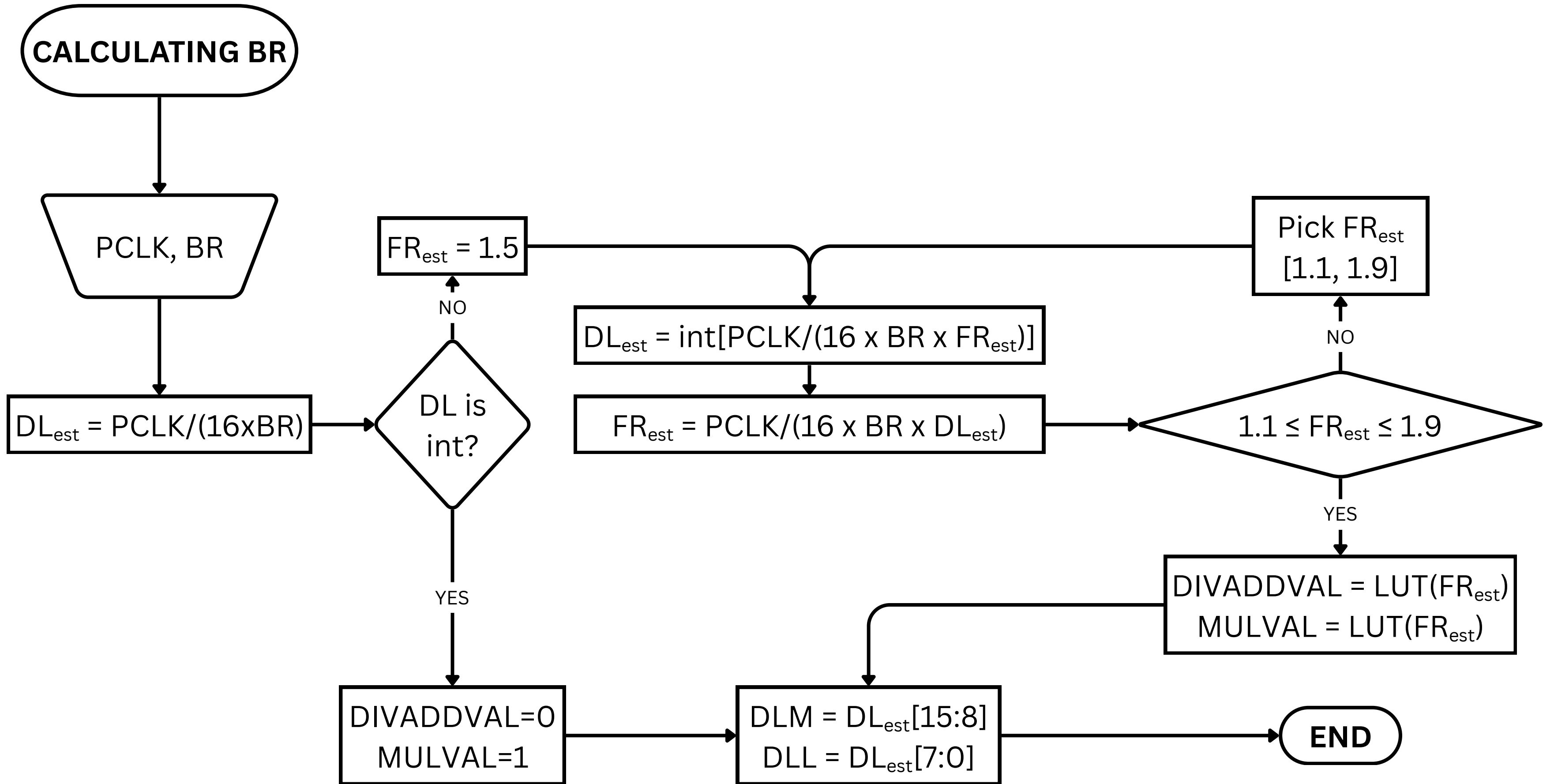


Table 287. Fractional Divider setting look-up table

FR	DivAddVal/ MulVal	FR	DivAddVal/ MulVal	FR	DivAddVal/ MulVal	FR	DivAddVal/ MulVal
1.000	0/1	1.250	1/4	1.500	1/2	1.750	3/4
1.067	1/15	1.267	4/15	1.533	8/15	1.769	10/13
1.071	1/14	1.273	3/11	1.538	7/13	1.778	7/9
1.077	1/13	1.286	2/7	1.545	6/11	1.786	11/14
1.083	1/12	1.300	3/10	1.556	5/9	1.800	4/5
1.091	1/11	1.308	4/13	1.571	4/7	1.818	9/11
1.100	1/10	1.333	1/3	1.583	7/12	1.833	5/6
1.111	1/9	1.357	5/14	1.600	3/5	1.846	11/13
1.125	1/8	1.364	4/11	1.615	8/13	1.857	6/7
1.133	2/15	1.375	3/8	1.625	5/8	1.867	13/15
1.143	1/7	1.385	5/13	1.636	7/11	1.875	7/8
1.154	2/13	1.400	2/5	1.643	9/14	1.889	8/9
1.167	1/6	1.417	5/12	1.667	2/3	1.900	9/10
1.182	2/11	1.429	3/7	1.692	9/13	1.909	10/11
1.200	1/5	1.444	4/9	1.700	7/10	1.917	11/12
1.214	3/14	1.455	5/11	1.714	5/7	1.923	12/13
1.222	2/9	1.462	6/13	1.727	8/11	1.929	13/14
1.231	3/13	1.467	7/15	1.733	11/15	1.933	14/15

PCLK = 14.7456 MHz

BR = 9600

$$DL_{\text{est}} = \text{PCLK} / (16 \times \text{BR})$$

$$DL_{\text{est}} = 14745600 / (16 \times 9600)$$

$$DL_{\text{est}} = 96$$

$$\text{DIVADDVAL} = 0$$

$$\text{MULVAL} = 1$$

$$\text{DLM} = 0$$

$$\text{DLL} = 96$$

PCLK = 12 MHz

BR = 115200

$$DL_{\text{est}} = 12000000 / (16 \times 115200)$$

$$DL_{\text{est}} = 6.51 \qquad FR_{\text{est}} = 1.5$$

$$DL_{\text{est}} = 12000000 / (16 \times 115200 \times 1.5)$$

$$DL_{\text{est}} = \text{int}[4.34] = 4$$

$$FR_{\text{est}} = 12000000 / (16 \times 115200 \times 4)$$

$$FR_{\text{est}} = 1.628$$

$$\text{DIVADDVAL} = \text{LUT}(1.625) = 5$$

$$\text{MULVAL} = \text{LUT}(1.625) = 8$$

$$\text{DLM} = 0$$

$$\text{DLL} = 4$$

$$\text{BR} = 115384$$

$$E_{\text{rel}} = 0.16 \%$$

14.4.1 UARTn Receiver Buffer Register (U0RBR - 0x4000 C000, U2RBR - 0x4009 8000, U3RBR - 0x4009 C000 when **DLAB = 0**)

The UnRBR is the top byte of the UARTn Rx FIFO. The top byte of the Rx FIFO contains the oldest character received and can be read via the bus interface. The LSB (bit 0) represents the “oldest” received data bit. If the character received is **less than 8 bits** the unused MSBs are **padding with zeroes**

The Divisor Latch Access Bit (DLAB) in LCR must be zero in order to access the UnRBR. The UnRBR is always read-only.

Since **PE, FE and BI bits correspond to the byte sitting on the top of the RBR FIFO** (i.e. the one that will be read in the next read from the RBR), the right approach for fetching the valid pair of received byte and its status bits is first to read the content of the U0LSR register, and then to read a byte from the UnRBR.

Table 272: UARTn Receiver Buffer Register (U0RBR - address 0x4000 C000, U2RBR - 0x4009 8000, U3RBR - 04009 C000 when DLAB = 0) bit description

Bit	Symbol	Description	Reset Value
7:0	RBR	The UARTn Receiver Buffer Register contains the oldest received byte in the UARTn Rx FIFO.	Undefined
31:8	-	Reserved, the value read from a reserved bit is not defined.	NA

14.4.2 UARTn Transmit Holding Register (U0THR - 0x4000 C000, U2THR - 0x4009 8000, U3THR - 0x4009 C000 when **DLAB = 0**)

The UnTHR is the top byte of the UARTn TX FIFO. The top byte is the newest character in the TX FIFO and can be written via the bus interface. The LSB represents the first bit to transmit.

The Divisor Latch Access Bit (DLAB) in UnLCR must be zero in order to access the UnTHR. The UnTHR is always write-only.

Table 273: UARTn Transmit Holding Register (U0THR - address 0x4000 C000, U2THR - 0x4009 8000, U3THR - 0x4009 C000 when DLAB = 0) bit description

Bit	Symbol	Description	Reset Value
7:0	THR	Writing to the UARTn Transmit Holding Register causes the data to be stored in the UARTn transmit FIFO. The byte will be sent when it reaches the bottom of the FIFO and the transmitter is available.	NA
31:8	-	Reserved, user software should not write ones to reserved bits.	NA

Table 288: UARTn Transmit Enable Register (U0TER - address 0x4000 C030, U2TER - 0x4009 8030, U3TER - 0x4009 C030) bit description

Bit	Symbol	Description	Reset Value
6:0	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
7	TXEN	When this bit is 1, as it is after a Reset, data written to the THR is output on the TXD pin as soon as any preceding data has been sent. If this bit is cleared to 0 while a character is being sent, the transmission of that character is completed, but no further characters are sent until this bit is set again. In other words, a 0 in this bit blocks the transfer of characters from the THR or TX FIFO into the transmit shift register. Software implementing software-handshaking can clear this bit when it receives an XOFF character (DC3). Software can set this bit again when it receives an XON (DC1) character.	1

Table 279: UARTn FIFO Control Register (U0FCR - address 0x4000 C008, U2FCR - 0x4009 8008, U3FCR - 0x4009 C008) bit description

Bit	Symbol	Value	Description	Reset Value
0	FIFO Enable	0	UARTn FIFOs are disabled. Must not be used in the application.	0
		1	Active high enable for both UARTn Rx and TX FIFOs and UnFCR[7:1] access. This bit must be set for proper UART operation. Any transition on this bit will automatically clear the related UART FIFOs.	
1	RX FIFO Reset	0	No impact on either of UARTn FIFOs.	0
		1	Writing a logic 1 to UnFCR[1] will clear all bytes in UARTn Rx FIFO, reset the pointer logic. This bit is self-clearing.	
2	TX FIFO Reset	0	No impact on either of UARTn FIFOs.	0
		1	Writing a logic 1 to UnFCR[2] will clear all bytes in UARTn TX FIFO, reset the pointer logic. This bit is self-clearing.	
3	DMA Mode Select		When the FIFO enable bit (bit 0 of this register) is set, this bit selects the DMA mode. See Section 14.4.6.1 .	0
5:4	-	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
7:6	RX Trigger Level		These two bits determine how many receiver UARTn FIFO characters must be written before an interrupt or DMA request is activated.	0
		00	Trigger level 0 (1 character or 0x01)	
		01	Trigger level 1 (4 characters or 0x04)	
		10	Trigger level 2 (8 characters or 0x08)	
		11	Trigger level 3 (14 characters or 0x0E)	
31:8	-		Reserved, user software should not write ones to reserved bits.	NA

Table 281: UARTn Line Status Register (U0LSR - address 0x4000 C014, U2LSR - 0x4009 8014, U3LSR - 0x4009 C014)
bit description

Bit	Symbol	Value	Description	Reset Value
0	Receiver Data Ready (RDR)		UnLSR0 is set when the UnRBR holds an unread character and is cleared when the UARTn RBR FIFO is empty.	0
		0	The UARTn receiver FIFO is empty.	
		1	The UARTn receiver FIFO is not empty.	
1	Overrun Error (OE)		The overrun error condition is set as soon as it occurs. An UnLSR read clears UnLSR1. UnLSR1 is set when UARTn RSR has a new character assembled and the UARTn RBR FIFO is full. In this case, the UARTn RBR FIFO will not be overwritten and the character in the UARTn RSR will be lost.	0
		0	Overrun error status is inactive.	
		1	Overrun error status is active.	
2	Parity Error (PE)		When the parity bit of a received character is in the wrong state, a parity error occurs. An UnLSR read clears UnLSR[2]. Time of parity error detection is dependent on UnFCR[0]. Note: A parity error is associated with the character at the top of the UARTn RBR FIFO.	0
		0	Parity error status is inactive.	
		1	Parity error status is active.	
3	Framing Error (FE)		When the stop bit of a received character is a logic 0, a framing error occurs. An UnLSR read clears UnLSR[3]. The time of the framing error detection is dependent on UnFCR0. Upon detection of a framing error, the Rx will attempt to resynchronize to the data and assume that the bad stop bit is actually an early start bit. However, it cannot be assumed that the next received byte will be correct even if there is no Framing Error. Note: A framing error is associated with the character at the top of the UARTn RBR FIFO.	0
		0	Framing error status is inactive.	
		1	Framing error status is active.	

Table 281: UARTn Line Status Register (U0LSR - address 0x4000 C014, U2LSR - 0x4009 8014, U3LSR - 0x4009 C014)
bit description

Bit	Symbol	Value	Description	Reset Value
4	Break Interrupt (BI)		When RXDn is held in the spacing state (all zeroes) for one full character transmission (start, data, parity, stop), a break interrupt occurs. Once the break condition has been detected, the receiver goes idle until RXDn goes to marking state (all ones). An UnLSR read clears this status bit. The time of break detection is dependent on UnFCR[0]. Note: The break interrupt is associated with the character at the top of the UARTn RBR FIFO.	0
		0	Break interrupt status is inactive.	
		1	Break interrupt status is active.	
5	Transmitter Holding Register Empty (THRE))		THRE is set immediately upon detection of an empty UARTn THR and is cleared on a UnTHR write.	1
		0	UnTHR contains valid data.	
		1	UnTHR is empty.	
6	Transmitter Empty (TEMT)		TEMT is set when both UnTHR and UnTSR are empty; TEMT is cleared when either the UnTSR or the UnTHR contain valid data.	1
		0	UnTHR and/or the UnTSR contains valid data.	
		1	UnTHR and the UnTSR are empty.	
7	Error in RX FIFO (RXFE)		UnLSR[7] is set when a character with a Rx error such as framing error, parity error or break interrupt, is loaded into the UnRBR. This bit is cleared when the UnLSR register is read and there are no subsequent errors in the UARTn FIFO.	0
		0	UnRBR contains no UARTn RX errors or UnFCR[0]=0.	
		1	UARTn RBR contains at least one UARTn RX error.	
31:8	-		Reserved, the value read from a reserved bit is not defined.	NA

Table 276: UARTn Interrupt Enable Register (U0IER - address 0x4000 C004, U2IER - 0x4009 8004, U3IER - 0x4009 C004 when DLAB = 0) bit description

Bit	Symbol	Value	Description	Reset Value
0	RBR Interrupt Enable		Enables the Receive Data Available interrupt for UARTn. It also controls the Character Receive Time-out interrupt.	0
		0	Disable the RDA interrupts.	
		1	Enable the RDA interrupts.	
1	THRE Interrupt Enable		Enables the THRE interrupt for UARTn. The status of this can be read from UnLSR[5].	0
		0	Disable the THRE interrupts.	
		1	Enable the THRE interrupts.	
2	RX Line Status Interrupt Enable		Enables the UARTn RX line status interrupts. The status of this interrupt can be read from UnLSR[4:1].	0
		0	Disable the RX line status interrupts.	
		1	Enable the RX line status interrupts.	
7:3	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
8	ABEOIntEn		Enables the end of auto-baud interrupt.	0
		0	Disable end of auto-baud Interrupt.	
		1	Enable end of auto-baud Interrupt.	
9	ABTOIntEn		Enables the auto-baud time-out interrupt.	0
		0	Disable auto-baud time-out Interrupt.	
		1	Enable auto-baud time-out Interrupt.	
31:10	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

Table 277: UARTn Interrupt Identification Register (U0IIR - address 0x4000 C008, U2IIR - 0x4009 8008, U3IIR - 0x4009 C008) bit description

Bit	Symbol	Value	Description	Reset Value
0	IntStatus		Interrupt status. Note that UnIIR[0] is active low. The pending interrupt can be determined by evaluating UnIIR[3:1].	1
		0	At least one interrupt is pending.	
		1	No interrupt is pending.	
3:1	IntId		Interrupt identification. UnIER[3:1] identifies an interrupt corresponding to the UARTn Rx or TX FIFO. All other combinations of UnIER[3:1] not listed below are reserved (000,100,101,111).	0
		011	1 - Receive Line Status (RLS).	
		010	2a - Receive Data Available (RDA).	
		110	2b - Character Time-out Indicator (CTI).	
		001	3 - THRE Interrupt	
5:4	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
7:6	FIFO Enable		Copies of UnFCR[0].	0
8	ABEOInt		End of auto-baud interrupt. True if auto-baud has finished successfully and interrupt is enabled.	0
9	ABTOInt		Auto-baud time-out interrupt. True if auto-baud has timed out and interrupt is enabled.	0
31:10	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

Table 278: UARTn Interrupt Handling

U0IIR[3:0] value ^[1]	Priority	Interrupt Type	Interrupt Source	Interrupt Reset
0001	-	None	None	-
0110	Highest	RX Line Status / Error	OE ^[2] or PE ^[2] or FE ^[2] or BI ^[2]	UnLSR Read ^[2]
0100	Second	RX Data Available	Rx data available or trigger level reached in FIFO (UnFCR0=1)	UnRBR Read ^[3] or UARTn FIFO drops below trigger level
1100	Second	Character Time-out indication	Minimum of one character in the Rx FIFO and no character input or removed during a time period depending on how many characters are in FIFO and what the trigger level is set at (3.5 to 4.5 character times). The exact time will be: $[(\text{word length}) \times 7 - 2] \times 8 + [(\text{trigger level} - \text{number of characters}) \times 8 + 1] \text{ RCLKs}$	UnRBR Read ^[3]
0010	Third	THRE	THRE ^[2]	UnIIR Read (if source of interrupt) or THR write ^[4]

Table 283: UARTn Auto-baud Control Register (U0ACR - address 0x4000 C020, U2ACR - 0x4009 8020, U3ACR - 0x4009 C020) bit description

Bit	Symbol	Value	Description	Reset value
0	Start		This bit is automatically cleared after auto-baud completion.	0
		0	Auto-baud stop (auto-baud is not running).	
		1	Auto-baud start (auto-baud is running). Auto-baud run bit. This bit is automatically cleared after auto-baud completion.	
1	Mode		Auto-baud mode select bit.	0
		0	Mode 0.	
		1	Mode 1.	
2	AutoRestart	0	No restart.	0
		1	Restart in case of time-out (counter restarts at next UARTn Rx falling edge)	0
7:3	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
8	ABEOIntClr		End of auto-baud interrupt clear bit (write-only accessible). Writing a 1 will clear the corresponding interrupt in the UnIIR. Writing a 0 has no impact.	0
9	ABTOIntClr		Auto-baud time-out interrupt clear bit (write-only accessible). Writing a 1 will clear the corresponding interrupt in the UnIIR. Writing a 0 has no impact.	0
31:10	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

Table 282: UARTn Scratch Pad Register (U0SCR - address 0x4000 C01C, U2SCR - 0x4009 801C, U3SCR - 0x4009 C01C) bit description			
Bit	Symbol	Description	Reset Value
7:0	Pad	A readable, writable byte.	0x00
31:8	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

Table 284: UARTn IrDA Control Register (U0ICR - 0x4000 C024, U2ICR - 0x4009 8024, U3ICR - 0x4009 C024) bit description

Bit	Symbol	Value	Description	Reset value
0	IrDAEn	0	IrDA mode on UARTn is disabled, UARTn acts as a standard UART.	0
		1	IrDA mode on UARTn is enabled.	
1	IrDAInv		When 1, the serial input is inverted. This has no effect on the serial output. When 0, the serial input is not inverted.	0
2	FixPulseEn		When 1, enabled IrDA fixed pulse width mode.	0
5:3	PulseDiv		Configures the pulse when FixPulseEn = 1. See text below for details.	0
31:6	-	NA	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	0

Table 285: IrDA Pulse Width

FixPulseEn	PulseDiv	IrDA Transmitter Pulse width (μ s)
0	x	$3 / (16 \times \text{baud rate})$
1	0	$2 \times T_{PCLK}$
1	1	$4 \times T_{PCLK}$
1	2	$8 \times T_{PCLK}$
1	3	$16 \times T_{PCLK}$
1	4	$32 \times T_{PCLK}$
1	5	$64 \times T_{PCLK}$
1	6	$128 \times T_{PCLK}$
1	7	$256 \times T_{PCLK}$

LPC_UARTx →

```
typedef struct {  
    union {  
        __I uint8_t RBR;  
        __O uint8_t THR;  
        __IO uint8_t DLL;  
        uint32_t RESERVED0;  
    };  
    union {  
        __IO uint8_t DLM;  
        __IO uint32_t IER;  
    };  
    union {  
        __I uint32_t IIR;  
        __O uint8_t FCR;  
    };  
};
```

```
    __IO uint8_t LCR;  
    uint8_t RESERVED1[7];  
    __I uint8_t LSR;  
    uint8_t RESERVED2[7];  
    __IO uint8_t SCR;  
    uint8_t RESERVED3[3];  
    __IO uint32_t ACR;  
    __IO uint8_t ICR;  
    uint8_t RESERVED4[3];  
    __IO uint8_t FDR;  
    uint8_t RESERVED5[7];  
    __IO uint8_t TER;  
    uint8_t RESERVED6[39];  
    __IO uint32_t FIFOLVL;  
} LPC_UART_TypeDef;
```

DRIVERS

```
typedef struct {
    uint32_t Baud_rate;          /**< UART baud rate */
    UART_PARITY_Type Parity;     /**< Parity selection, should be:
    - UART_PARITY_NONE: No parity
    - UART_PARITY_ODD: Odd parity
    - UART_PARITY_EVEN: Even parity
    - UART_PARITY_SP_1: Forced "1" stick parity
    - UART_PARITY_SP_0: Forced "0" stick parity
    */
    UART_DATABIT_Type Databits;  /**< Number of data bits, should be:
    - UART_DATABIT_5: UART 5 bit data mode
    - UART_DATABIT_6: UART 6 bit data mode
    - UART_DATABIT_7: UART 7 bit data mode
    - UART_DATABIT_8: UART 8 bit data mode
    */
    UART_STOPBIT_Type Stopbits;  /**< Number of stop bits, should be:
    - UART_STOPBIT_1: UART 1 Stop Bits Select
    - UART_STOPBIT_2: UART 2 Stop Bits Select
    */
} UART_CFG_Type;
```



```
typedef struct {  
    FunctionalState FIFO_ResetRxBuf;    /**< Reset Rx FIFO command state , should be:  
        - ENABLE: Reset Rx FIFO in UART  
        - DISABLE: Do not reset Rx FIFO in UART  
    */  
  
    FunctionalState FIFO_ResetTxBuf;    /**< Reset Tx FIFO command state , should be:  
        - ENABLE: Reset Tx FIFO in UART  
        - DISABLE: Do not reset Tx FIFO in UART  
    */  
  
    FunctionalState FIFO_DMAMode;       /**< DMA mode, should be:  
        - ENABLE: Enable DMA mode in UART  
        - DISABLE: Disable DMA mode in UART  
    */  
  
    UART_FIT0_LEVEL_Type FIFO_Level;    /**< Rx FIFO trigger level, should be:  
        - UART_FIFO_TRGLEV0: UART FIFO trigger level 0: 1 character  
        - UART_FIFO_TRGLEV1: UART FIFO trigger level 1: 4 character  
        - UART_FIFO_TRGLEV2: UART FIFO trigger level 2: 8 character  
        - UART_FIFO_TRGLEV3: UART FIFO trigger level 3: 14 character  
    */  
  
} UART_FIFO_CFG_Type;
```



```
void UART_Init(LPC_UART_TypeDef *UARTx, UART_CFG_Type *UART_ConfigStruct);  
void UART_DeInit(LPC_UART_TypeDef* UARTx);  
void UART_ConfigStructInit(UART_CFG_Type *UART_InitStruct);
```

```
void UART_SendByte(LPC_UART_TypeDef* UARTx, uint8_t Data);  
uint8_t UART_ReceiveByte(LPC_UART_TypeDef* UARTx);  
uint32_t UART_Send(LPC_UART_TypeDef *UARTx, uint8_t *txbuf, uint32_t buflen, TRANSFER_BLOCK_Type flag);  
uint32_t UART_Receive(LPC_UART_TypeDef *UARTx, uint8_t *rxbuf, uint32_t buflen, TRANSFER_BLOCK_Type flag);
```

```
void UART_FIFOConfig(LPC_UART_TypeDef *UARTx, UART_FIFO_CFG_Type *FIFOCfg);  
void UART_FIFOConfigStructInit(UART_FIFO_CFG_Type *UART_FIFOInitStruct);
```

```
uint32_t UART_GetIntId(LPC_UART_TypeDef* UARTx);  
uint8_t UART_GetLineStatus(LPC_UART_TypeDef* UARTx);
```

```
void UART_IntConfig(LPC_UART_TypeDef *UARTx, UART_INT_Type UARTIntCfg, FunctionalState NewState);  
void UART_TxCmd(LPC_UART_TypeDef *UARTx, FunctionalState NewState);  
FlagStatus UART_CheckBusy(LPC_UART_TypeDef *UARTx);  
void UART_ForceBreak(LPC_UART_TypeDef* UARTx);
```

```
void UART_ABClearIntPending(LPC_UART_TypeDef *UARTx, UART_ABE0_Type ABIntType);  
void UART_ABCmd(LPC_UART_TypeDef *UARTx, UART_AB_CFG_Type *ABConfigStruct, FunctionalState NewState);
```